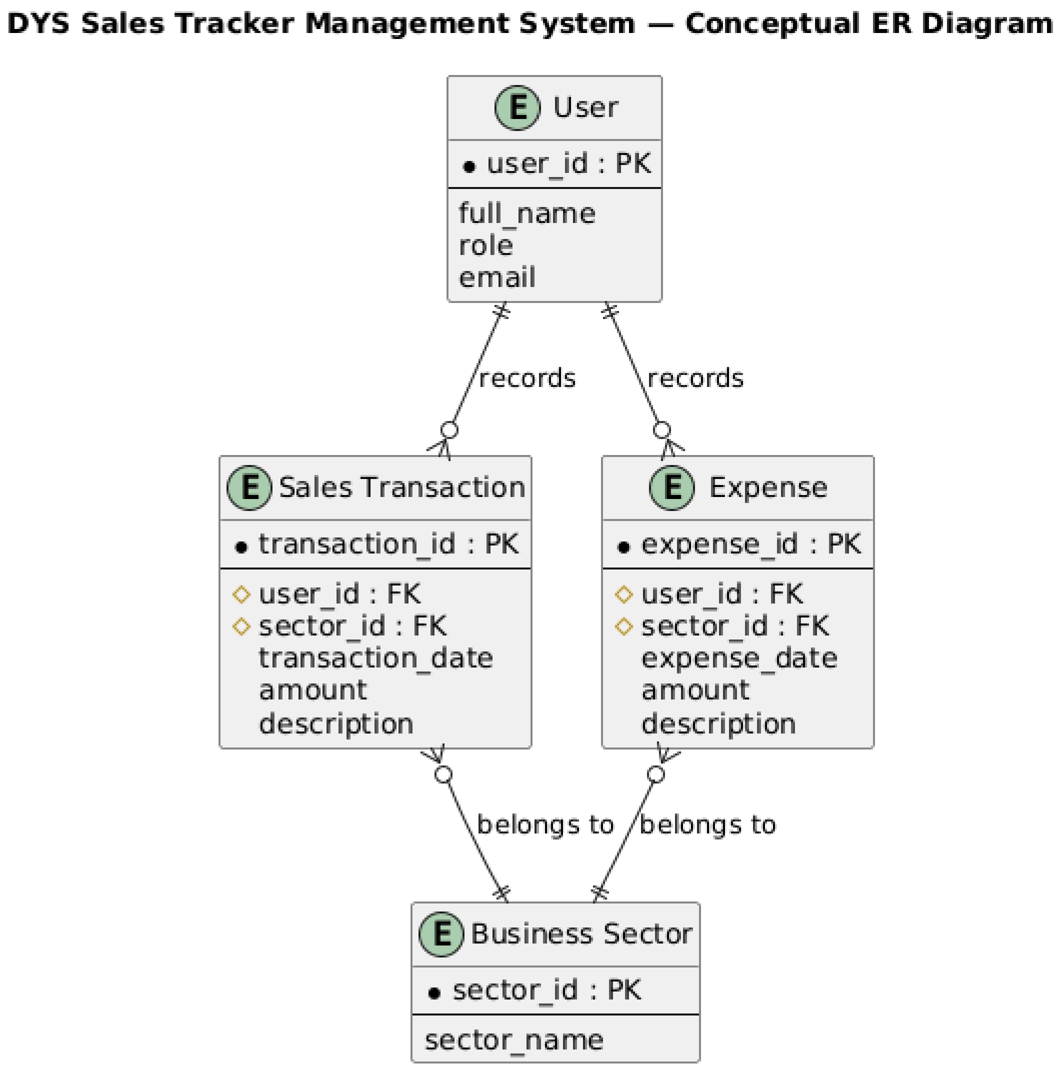

(ER) Diagram Documentation

1. Purpose

The Entity Relationship (ER) Diagram provides the conceptual foundation for the data structure of the DYS Sales Tracker Management System. It identifies the core business entities required to support centralized financial transaction monitoring across the four business sectors of DYS Events Management, and defines how these entities relate to one another. This conceptual model ensures that data storage requirements are clearly understood before implementation-level design decisions are made and serves as the foundation for the logical database schema of the system.

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2. Entity Summary

Entity	Description	Purpose
User	Represents an individual who interacts with the system under one of three roles: Business Owner, Event Manager, or Employee/Event Staff.	Supports authentication and role-based access to system functions.
Business Sector	Represents one of the four business divisions of DYS Events Management.	Supports the Business Sector Switcher feature by organizing financial data per division.
Sales Transaction	Represents a recorded sale entered into the system.	Supports sales recording and automated financial calculation.
Expense	Represents a recorded business expense entered into the system.	Supports expense recording and automated financial calculation.

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3. Relationship Summary

Relationship	Cardinality	Description
User records Sales Transaction	1:M	A single user may record many sales transactions over time.
User records Expense	1:M	A single user may record many expenses over time.
Sales Transaction belongs to Business Sector	M:1	Each sales transaction is associated with exactly one business sector.
Expense belongs to Business Sector	M:1	Each expense is associated with exactly one business sector.

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4. Design Decisions

Payroll is not modeled as an entity. Payroll is treated as an automated calculation derived from recorded transaction data. Since it is generated by the system rather than maintained as persistent business data, it is modeled as a system process instead of a database entity.

Booking, Client, and Event entities are not included. The conceptual database focuses only on the data required to support the approved system functions, specifically recording and monitoring sales transactions and business expenses. Therefore, additional entities outside this scope are not included.

No direct User–Business Sector relationship is modeled. Business sector access is represented through the application's role-based access mechanism. Since no fixed assignment between users and business sectors is required in the conceptual data model, a direct relationship is not included.

Scope Compliance. The ER Diagram includes only the entities and relationships necessary to support the approved system requirements. This ensures that the conceptual database model remains focused, consistent, and free from unnecessary complexity.

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5. Traceability to the Concept Paper

Entity	Supported Feature	Concept Paper Basis
User	Role-Based Access Control	Target User table defines Business Owner, Event Manager, and Employees/Event Staff as system users with differentiated access.
Business Sector	Business Sector Switcher	The Concept Paper states the system will allow the owner to switch between different business sectors, keeping records organized and separated.
Sales Transaction	Sales Recording & Automated Financial Calculator	The Concept Paper describes recording daily transactions and an automatic calculator that computes total sales.
Expense	Expense Recording & Automated Financial Calculator	The Concept Paper describes recording business expenses and the same automatic calculator computing totals.
